



Beyond Per Capita: Superlinear Scaling and Industrial Decoupling in Urban Water Metabolism

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Abstract

Traditional urban water management often relies on aggregate per capita indicators that fail to capture the complex dynamics and systemic inefficiencies of growing cities. This study applies Urban Scaling Theory to analyze consumptive water use across 343 Chilean communes, evaluating how demographic, environmental, and socioeconomic variables modulate metabolic performance in a context of chronic water security challenges. Our findings reveal a critical superlinear scaling of domestic consumption ($\beta \approx 1.08$), where water demand grows faster than population size, indicating that current urbanization patterns generate diseconomies of scale rather than expected infrastructure efficiencies. Conversely, industrial water use is found to be entirely decoupled from population size, driven instead by resource proximity in low-density territories. This reveals a functional and spatial mismatch where wealthy residential hubs consume water disproportionately while the environmental costs of industrial production are offshored to less affluent areas, commonly known as “sacrifice areas”. We conclude 1 that sustainable urban planning must move beyond simple per capita targets to address the sector-specific scaling properties and underlying socioeconomic inequities that govern urban water metabolism. Effective governance requires multi-scale strategies that integrate administrative management with biophysical constraints to ensure long-term water security and social equity.

Keywords Urban scaling · Water metabolism · Sustainable urban planning · Water consumption · Chile

1 Introduction

The Holocene’s climatic stability fostered the development of agriculture and complex human societies. However, anthropogenic pressures have destabilized the vital Earth systems upon which these societies depend. The Stockholm Resilience Institute quantified this risk by identifying nine “planetary boundaries”—critical thresholds for Earth’s system pro-

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cesses. This framework provides a concrete definition of sustainability, linking this goal directly to the challenge of remaining within the “safe and just operating space” defined by these boundaries (Rockström et al. 2009).

Sustainability itself must be grounded in the fundamental, non-negotiable laws of the natural world, as human and ecological systems are governed by the same physical conservation principles (Burger et al. 2012). This biophysical grounding requires recognizing that social and biological systems are hierarchical, nested, and interdependent, creating a complex interplay between local and global sustainability. Furthermore, all systems are fundamentally constrained by their regeneration capacity. These principles establish a foundational requirement for human societies to manage their ecological footprint and growth, operating within the biophysical limits of a finite planet.

Urbanization is a key anthropogenic pressure on the stability of Earth’s systems and a central challenge to achieving sustainability, both locally and globally. On a global scale, cities are responsible for 70% of greenhouse gas emissions due to direct and indirect impacts related to urban functioning, including intensive agriculture and fossil fuel use (UN-Habitat 2022; Burger et al. 2019; Delgado-Ramos 2015; Elmqvist et al. 2013; Güneralp et al. 2013). Locally, urban areas lead to environmental impacts due to temperature increases, habitat fragmentation, and the introduction of exotic species (Alberti and Marzluff 2004; Peng et al. 2020). These dynamics operate within co-dependent, hierarchical open systems. Consequently, the persistent growth of urban populations necessitates a deeper understanding of city dynamics to pursue sustainability across local and global scales.

Freshwater consumption is a critical concern directly linked to the planetary boundary framework and the need to operate within Earth’s regenerative capacity. While balancing human demand with environmental replenishment is precarious, it is essential for achieving global targets like the United Nations’ Sustainable Development Goal 6 (SDG 6). Access to water and sanitation—recognized as a human right in 2010—is foundational to life, equity, and socioeconomic development (UNGA 2010; UN-Water 2013; UN-Habitat 2022). However, recent projections indicate that water consumption may exceed the global freshwater planetary boundary between 2030 and 2050, exposing 9% of the global population to extreme scarcity, with urban areas being particularly vulnerable (Van Vuuren et al. 2025; Ravinandrasana and Franzke 2025). While public health and social justice discussions often prioritize access, the reciprocal impact of urbanization on water systems is equally critical. Urban development drives habitat loss, resource intensification, and waste generation (Burger et al. 2019; Loreau 2022; Guetté et al. 2017; Brondizio et al. 2019; Newbold et al. 2015), deteriorating water quality and availability through both direct mechanisms (e.g., groundwater overextraction) and indirect pressures (e.g., agricultural intensification).

To address the challenges posed by urban growth and resource consumption, as required by the first sustainability principle, a formal approach is needed to quantify urban dynamics. Water fluxes involved in urban processes and dynamics can be estimated based on instrumental measurements, variable extrapolation, or more complex models. Multiple theoretical frameworks underpin the assessment of water demand, availability, and even valuation (Babel et al. 2007; Nations 2021). One such approach is Urban Science, which examines the dynamics and patterns of cities at a global scale (Lobo et al. 2020; Batty 2013).

Within this domain, urban metabolism (UM) framework aims to assess metabolic flows and processes involved in the operation and growth of human settlements as complex systems. Particularly, the urban scaling theory (UST) allows us to examine how urban proper-

ties evolve with city size (N), typically modeled as the power-law relationship $Y=Y_0N^\beta$. Here, Y_0 is a normalization constant, and the scaling exponent β serves as a critical descriptor of the system's inherent dynamics. Empirically, this power-law is consistent in urban systems with different characteristics (universality), showing patterns suggesting that fundamental principles govern growth and organization (Bettencourt et al. 2007; Lobo et al. 2020; D'Acci 2024).

This relationship manifests in three distinct regimes. First, linear scaling ($\beta \approx 1$) indicates direct proportionality, typically observed in variables linked to basic individual needs, such as household water consumption. Second, superlinear scaling ($\beta > 1$) reflects increasing returns to scale—common in socioeconomic variables like productivity and crime—attributed to intensified agglomeration effects driven by social interactions. Finally, sublinear scaling ($\beta < 1$) represents economies of scale, where larger cities require proportionally fewer resources per unit of growth. This efficiency, often described as 'doing more with less,' is characteristic of infrastructural networks and biological systems operating under optimization constraints (Ribeiro and Rybski 2023).

While the strict universality of these laws is debated (Arcaute et al. 2015; Keuschnigg 2019a, b), the UST framework remains a compelling tool for the identification of metabolic patterns that deviate from expected efficiency in resource use (Alves et al. 2013; Liu et al. 2025). By establishing a baseline for urban scaling, UST provides a quantitative basis to detect inefficiencies and understand long-term systemic sustainability (Bettencourt 2013; Sarkar 2019; Burger et al. 2022; Cottineau et al. 2016). Chile presents a unique case study for applying this framework due to its distinct geographic, demographic, and hydrological configuration. Bounded by the Pacific Ocean to the west and the Andes Mountains to the east, the continental territory spans over 4,270 km, exhibiting a pronounced latitudinal precipitation gradient (Morrone 2015). At the national scale, the economy is dominated by mining, manufacturing, and the silvo-agricultural sector, all of which exert substantial and spatially heterogeneous pressures on water demand (Central Bank of Chile 2024).

The country hosts a population exceeding 18 million, with over 87% concentrated in urban areas (INE 2018, 2025). These demographic pressures operate within a water governance framework defined by the Water Code, where rights are private and tradable—a system marked by high institutional fragmentation that often hinders effective management (Congreso Nacional de Chile 1981; Valdés-Pineda et al. 2014). Furthermore, the hydrological scenario has deteriorated significantly over the last decade; numerous river basins face high to extreme water stress driven by a 'megadrought', with estimates suggesting that sustainable allocation limits in northern and central regions have already been surpassed (Alvarez-Garretón et al. 2023; Marquet et al. 2022).

The objective of this study is to analyze the influence of demographic, geographic, meteorological, and socioeconomic factors on urban water consumption across Chilean communes—the fundamental administrative unit for governance and data in the absence of a unified functional definition of a city. Using UST, we assess the relationship between size and water use, identifying significant deviations from expected scaling behaviors. Our findings deliver a critical, quantitative basis for comprehending water security determinants, informing policy and management decisions aimed at sustainable urban development in rapidly densifying environments.

To our knowledge, this is the first application of Urban Scaling Theory to consumptive water use across an entire national urban system in Latin America. Three contribu-

tions distinguish this work: (i) we demonstrate that domestic water use in Chile follows a robust superlinear scaling regime ($\beta > 1$) that persists across multiple model specifications and population-stratifications, thereby challenging the per-capita assumption embedded in conventional water demand planning; (ii) we document a structural decoupling between industrial water use and population size, revealing a “fiscal-functional mismatch” by which administrative tax-based statistics misrepresent the spatial distribution of environmental burden; and (iii) we identify a “paradox of density” in which population density is positively associated with domestic but negatively with total water use, with direct implications for densification policies.

2 Materials

This research employs a complementary methodological approach, combining cross-sectional analysis with allometric modeling, fundamentally rooted in the UM framework (Burger et al. 2022). We analyze water use dynamics at the commune level, the relevant administrative unit in the Chilean context. The subsequent sections detail (§2.1) the construction of an integrated socio-ecological dataset, (§3.1) the exploratory data analysis, and the application of both water use models: (§3.2) multivariate and (§3.3) size scaling. This dual approach allows us to explicitly connect the analysis of material flows (metabolism) with the systemic properties and resource efficiency of urban systems (scaling), characterizing both the localized drivers and the macroscopic, size-dependent laws of water consumption.

2.1 Dataset

2.1.1 Water Use Data

Annual water use series were obtained from the CR2WU database, which provides volumetric estimates of actual water use independent of legally granted water rights (Juan Pablo Boisier et al. 2024). This database uses two calculation methodologies: one for water use relating to land use sectors (Land Water Use, or LWU, available annually from 1960 to 2020), and one for all other sectors (Non-Land Water Use, or NLWU, available from 1950 to 2020). LWU estimates are derived primarily from surface water budgets for land use, land-use change, and forestry. NLWU is estimated using spatially explicit activity drivers combined with sector-specific consumption rates, covering drinking water, mining, energy, manufacturing, and livestock—each with associated subsectors. For drinking water (DW) specifically, the primary driver is communal population, with consumption rates derived from utility billing records for urban areas and per-capita demand coefficients for rural zones. Database outputs were validated by J. P. Boisier et al. (2025) against independent estimates from FAO and the National Water Directorate (DGA).

Consistent with our focus on consumptive blue water use, we excluded non-consumptive sectors such as hydroelectric generation and rainfed agriculture from the analysis. Consequently, we aggregated water uses into three distinct functional categories for this study (see Supplementary Information §1). The first is LWU, which is restricted here to consumptive withdrawals for irrigated agriculture and forestry. The second is DW, encompassing total

potable water consumption across both urban and rural sectors. Finally, the Industrial category aggregates the remaining productive sectors. This classification allows us to assess the distinct metabolic characteristics of direct human consumption, land-dependent production, and non-land-based industrial activities.

2.1.2 Demographic, Environmental, and Socioeconomic Data

Our analysis seeks to characterize the distinct social, ecological, and economic dimensions that influence urban metabolic processes and define the scale of the commune. Based on data availability, we compiled eleven predictor variables to capture the specific demographic, socioeconomic, and abiotic environmental profiles of the study area (more details in Supplementary Information §2).

Demographic: Total population (Pop) and population density (Density) were compiled from National Institute of Statistics records (2002–2020) (INE 2021; MINDESO 1992).

Environmental: Climate data were derived from the CR2MET gridded dataset (0.05°) (Juan Pablo Boisier 2023). We calculated area-weighted communal means for Annual Mean Temperature (Temp) and Total Precipitation (Precip) using the terra R package (Hijmans 2025). Latitude (Lat) was included to represent the country's climatic and geographic gradient (17–55°S).

Socioeconomic: We utilized the Socio-Material Territorial Index (ISMT) (OCUC 2019) as a proxy for socioeconomic status or household wealth. This index was computed by OCUC using 2017 census data. Productive capacity was characterized using tax declaration data from 2017 to maintain consistency with the ISMT variable, yielding the number of firms (Firms), total revenue (Revenue), and dependent employees (Employees) (SII 2024). Additionally, the resident workforce (Workers) was obtained from the CASEN survey (CASEN 2017).

The selected variables represent the core dimensions that drive, constrain, or modulate the water metabolism of the communes. Demographic data (Pop, Density) define the scale of urban organization, a central element in UST. Environmental variables (Temp, Precip, Lat) act as biophysical constraints on metabolism, reflecting the available climate context and natural resource base. Finally, the socioeconomic and productive indicators (ISMT, Firms, Revenue, Employees, Rev/firm, Workers) characterize the functional structure and complexity of the commune, influencing the intensity and pattern of metabolic flows.

3 Methods

3.1 Data Analysis

We examined water consumption trends from 1950 to 2020 using locally weighted scatterplot smoothing (LOWESS) regression to visualize aggregate patterns. Sectoral distributions and regional variations were also analyzed relative to population data.

The data analysis is structured around two complementary research questions: (a) What are the local and short-term determinants that explain variations in water use across heterogeneous landscapes? and (b) Do universal macroscopic laws describe how water use scales with population size? To address (a), we employed a multivariable model to identify the

relative contribution of localized drivers (climate, economy, demographics). For (b), we utilized UST to evaluate scaling, revealing potentially systemic efficiency or inefficiency.

An exploratory data analysis was conducted to assess the distribution and suitability of all predictor variables. We addressed redundancy and multicollinearity by examining pairwise Spearman rank-correlations (ρ) and Variance Inflation Factors (VIF); variables showing high redundancy or VIF values were excluded. The underlying variance structure was further characterized using Principal Component Analysis (PCA). Linearity was assessed via scatter plots. To satisfy modeling assumptions, skewed variables were log-transformed, and all predictors were standardized using Z-scores to ensure scale consistency.

We modeled water use as the response variable using two complementary approaches. First, we constructed multivariate models incorporating the selected demographic, socio-economic, climatic, and geophysical predictors (Eq. 1). Second, we applied scaling (UST) to describe macroscopic consumption trends as a function of population size (Eq. 2).

3.2 Multivariate Modeling of Water Use

To identify the determinants of water consumption, we constructed a multivariate log-linear model (Eq. 1) using Ordinary least squares (OLS) fit. In this specification, the coefficients β represent the elasticity of water use with respect to the predictors (vector X):

$$\log(WU) = \beta_0 + \beta^T \log(X) \quad (1)$$

Given the periodicity of census-derived socioeconomic data, the model was applied to a set of communes for a single representative year, an approach known as cross-sectional or transversal scaling (Ribeiro et al. 2020). We addressed multicollinearity by removing redundant predictors ($\rho > 0.8$ or $VIF > 10$), after which a bidirectional stepwise regression determined the final set of explanatory variables. Separate linear regression models were then fitted for DW, the Industrial sector, and the sum of both. To identify the most parsimonious set of predictors, we employed a bidirectional stepwise selection procedure based on the Akaike Information Criterion (AIC) (Quinn and Keough 2002). Subsequently, model performance was validated through standard residual diagnostics, and the relative importance of each driver was inferred from the standardized coefficients.

3.3 Urban Scaling

We evaluated the allometric scaling of water consumption relative to population size. OLS regression was employed on log-transformed variables (Equation [eq:logwu]). Goodness-of-fit was assessed via the coefficient of determination (R^2) and residual analysis (Quinn and Keough 2002). The slope (β_1) estimates the scaling exponent (β), classifying the relationship into three distinct metabolic regimes: sublinear ($\beta < 1$, economies of scale), linear ($\beta \approx 1$, proportionality), or superlinear ($\beta > 1$, increasing returns to scale).

$$\log(WU) = \beta_0 + \beta_1 \log(Pop) \quad (2)$$

To complement the central tendency estimates provided by OLS, we employed quantile regression to characterize the scaling relationship across the conditional distribution of

water use. This approach allows for the assessment of potential heterogeneity in metabolic regimes, specifically by estimating scaling exponents for the lower ($\tau = 0.1$) and upper ($\tau = 0.9$) boundaries (Koenker and Bassett 1978). By distinguishing between these quantiles, we can identify divergent scaling patterns for communes with low versus high water consumption intensity relative to their population size.

Scaling patterns were analyzed for both total and urban populations, and for size-dependence stratified by a population threshold of 20,000 inhabitants. To address the absence of a standardized city definition, we distinguish between small towns and rural areas, and larger urban communes (MINVU 2009).

4 Results

4.1 Spatio-Temporal Characterization of Water Use

National water consumption has exhibited a consistent upward trend since 1950, punctuated by episodic peaks associated with agricultural demand (Fig. 1). Irrigated agriculture (LWU) consistently accounts for over 80% of national consumptive use (Fig. 1A). However, the composition of other water uses has shifted significantly over time: until the 1990s, DW was the predominant sector (>60%), after which industry, mining, and energy consumption experienced rapid expansion. Notably, mining water use increased by over 400% between

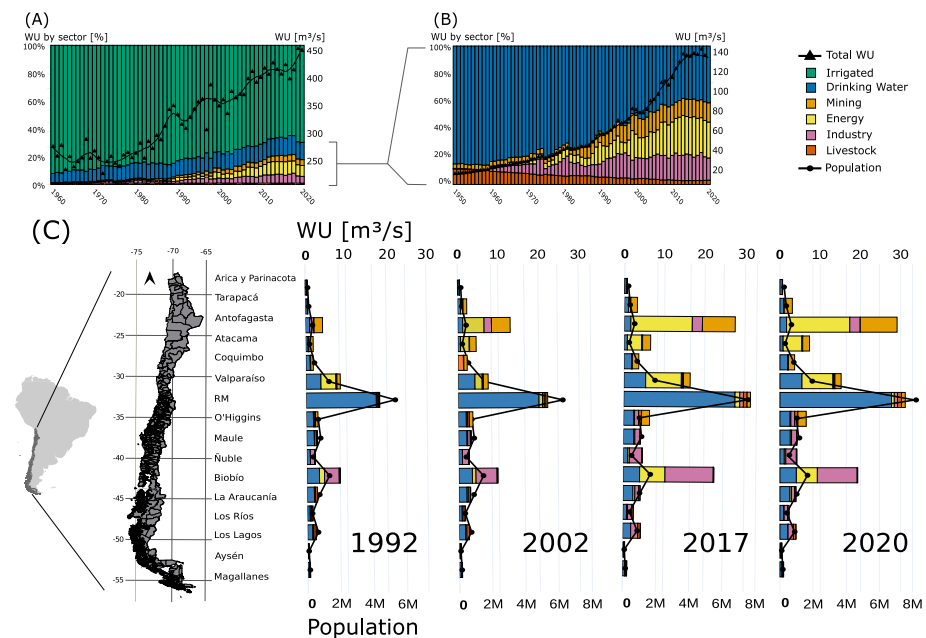


Fig. 1 National annual time series of consumptive water use (10-point simple moving average) for (A) total water consumption (LWU+DW+all Industrial sectors) and (B) water consumption excluding LWU (green). Stacked bars indicate the relative contribution of each sector (summing to 100%). (C) Regional population and sectoral use for reference years (excluding LWU)

1990 and 2020, a trend directly attributable to the sectoral growth of the mining activity (Fig. 1B).

Figure 1C summarizes regional and sectoral consumption alongside population for four reference years. Agriculture is excluded to visualize non-agricultural trends. Water use for energy and mining has intensified in the northern regions, particularly in Antofagasta, while industrial demand has driven growth primarily in the South-Central Biobio region. The Metropolitan Region (RM) remains distinguished by its high domestic water use, consistent with its dense population concentration ($\sim 7 \times 10^6$ inhabitants at 2020, representing ~ 40 % of the national population).

4.2 Exploratory Data Analysis of Model Variables

Table 1 presents the descriptive statistics for all parameters: the three response variables representing consumptive water use (DW, Industrial, and LWU), and the eleven potential predictor variables. These values correspond to data from each Chilean commune for a specific year, as socioeconomic quantification is based on the 2017 census, the most recent published record. Exploratory analysis revealed strongly right-skewed distributions of water use across communes, motivating variable transformation before modeling (see Supplementary Information §2).

In the multivariate dataset, none of the eleven predictors followed a normal distribution, and several displayed marked skewness. Furthermore, variables differed substantially in magnitude and units; for example, population is measured in thousands of inhabitants, whereas the ISMT is a normalized index (0–1). To address this heterogeneity, all variables

Table 1 Summary of the response and significant explanatory variables of the multivariate water use model. The mean (μ), standard deviation (σ), and range represent the data for all communes over the course of one year (2017)

Variable	Unit	μ	σ	Range [min–max]
DW	m ³ /s	0.17	0.29	[0.002–1.87]
Industrial	m ³ /s	0.26	1.02	0–13.41]
LWU	m ³ /s	0.79	1.25	0–7.62]
Pop	no. inhabitants	5.68×10^4	8.49×10^4	[1,218–604,744]
Density	no. inhabitants/ km ²	956	2,845	[0.26–17,484]
Precip	mm	983.41	746.80	[1.68–3,621.91]
Temp	°C	12.59	2.56	[3.48–16.38]
Lat	-	-	-	[-53.643–-18.428]
ISMT	-	0.49	0.07	[0.36–0.78]
Firms	no. firms	3,623	7,055	[84–64,926]
Revenue	USD	7.01×10^7	5.03×10^8	[1.71 × 10 ⁴ –8.02 × 10 ⁹]
Rev/firm	USD/no. firms	6.46×10^3	1.33×10^4	[1.66 × 10 ² –1.27 × 10 ⁵]
Employees	no. employees	2.84×10^4	1.02×10^5	[1.07 × 10 ³ –1.16 × 10 ⁶]
Workers	no. workers	284	419	[25–3,086]

were log-transformed and standardized. A $\log(1+x)$ transformation (*i.e.*, $\log(1+x)$) was applied to variables containing zero values (*e.g.*, industrial water use) to ensure numerical stability.

Principal Component Analysis (PCA) of the explanatory variables reveals a clear dichotomy between socioeconomic/demographic factors and environmental factors (Fig. 2A). This distinct clustering suggests that urban size and economic activity represent dimensions of variance that are structurally independent of climatic drivers.

Socioeconomic variables exhibit strong intercorrelations and are closely associated with population size (Fig. 2B). Communes with larger populations tend to possess higher business density, employment rates, and economic activity. This collinearity indicates potential redundancy among predictors, which guided our variable selection process.

No significant associations were observed between the predictor variables and LWU in log–log transformations (Fig. 2C). Consequently, we excluded this water use type from the multivariate model.

Based on correlation analysis, multicollinearity diagnostics, and variable-response relationships, we excluded predictors that were redundant, exhibited high multicollinearity (Table §1), or lacked theoretical justification (see Supplemental Material). The final multiple regression model included: Population, Density, ISMT, Revenue per firm, Precipitation, and Temperature.

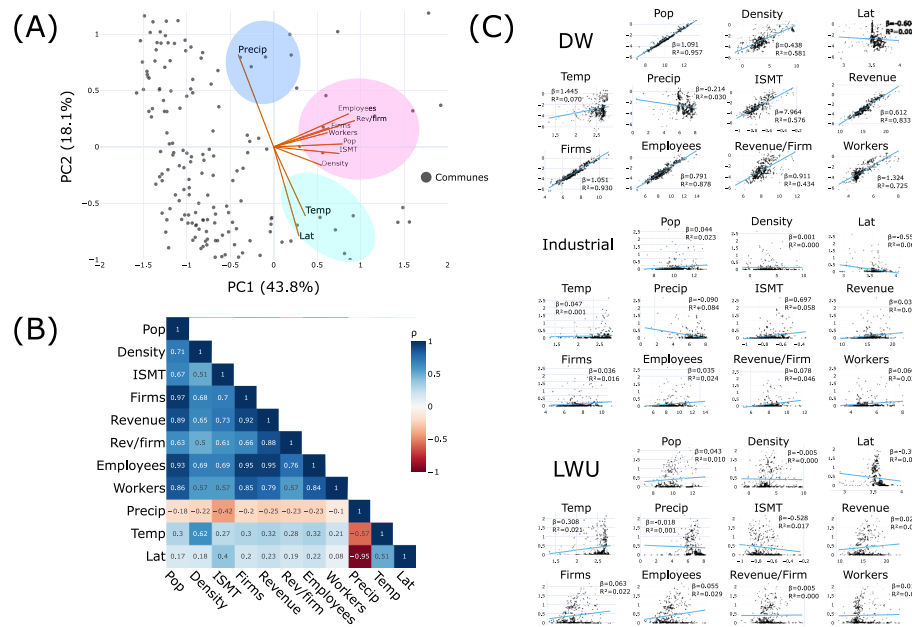


Fig. 2 Exploratory data analysis and variable selection for communal water use modeling. The unit of analysis for all panels represents Chilean communes. (A) Spearman rank correlation matrix (ρ) between the eleven explanatory variables. (B) Principal Component Analysis (PCA) Loading Plot for the explanatory variables. The first two principal components (PC1 and PC2) explain a total of 61.9% of the variance. The variables cluster into three distinct, underlying factors: (i) Socioeconomic/Demographic, (ii) Precipitation/Latitude, and (iii) Temperature. (C) Scatter plots illustrating the relationship between the three types of water consumption: DW, Industrial, and LWU against the candidate explanatory variables

4.3 Multivariate Modeling of Water Use

The NLWU model explains 67.2% of the variance. The residual standard error ($RSE=0.93$) suggests a moderate dispersion of predictions on the log scale. We observed stark differences in goodness-of-fit between the sub-sectors. While the model captures 98% of the variance for DW ($R_{adj}^2=0.98$, $F=4767$, $p<10^{-3}$), it explains only 12% of the variation in Industrial water use ($R_{adj}^2=0.12$, $F=12.68$, $p<10^{-3}$) (see Table 2).

NLWU is strongly influenced by population with a superlinear scaling ($\beta=1.14 \pm 0.09$), implying that communes twice as populous consume 220% more water, *ceteris paribus* (see details calculation in Supplementary Information §3).

Socioeconomic variables also function as key drivers. Sales per firm—a proxy for communal industrial productivity—shows a positive but modest effect ($\beta=0.13$, $p \sim 0.05$). While a positive coefficient aligns with the expectation that productive communes would use more water, the value <1 indicates sublinear scaling (economies of scale), where a 1% productivity increase yields a $<1\%$ rise in water use. Similarly, the ISMT indicates that consumption rises with commune-level wealth, though sublinearly ($\beta=0.35 \pm 0.07$).

Population density exhibits a significant negative coefficient ($\beta = -0.27$, $p \approx 1.5 \times 10^{-3}$), suggesting that higher density is associated with proportionally lower water use. This trend aligns with efficiency principles where denser communes occupy smaller areas, potentially reducing distribution losses. Furthermore, high-density areas are typically residential rather than industrial, contributing to lower overall volumetric demand.

Precipitation was the sole environmental predictor for NLWU, showing a negative relationship. Wetter communes likely require less water for outdoor activities; however, this effect was marginally significant in the DW model ($\beta = -0.01$, $p=0.36$). For industrial consumption, the lack of a clear pattern stems from the diversity of water sources (e.g., groundwater, surface, seawater). For instance, the highest industrial consumers (i.e., communes like Mejillones, Coronel, Puchuncaví) share a reliance on thermoelectric power generation—often using seawater for cooling—despite being located in distinct precipitation regimes.

The sector-specific models reveal a sharp contrast in performance. The DW model achieved an excellent fit ($R_{adj}^2=0.98$), with all predictors showing statistical significance ($p<0.001$). The population has a high influence on DW ($\beta=1.29 \pm 0.02$), which is expected

Table 2 Regression results of the multivariate log-linear (Cobb-Douglas) model for non-land water use (NLWU) in continental Chilean communes. Coefficients β represent partial elasticities of NLWU with respect to each predictor, holding the others constant; they are therefore not directly comparable to the bivariate scaling exponents reported in Fig. 3. $R_{adj}^2 = 0.672$, $F=140.6$, $p<10^{-3}$, $N=336$

Variable (log)	Coef. value β	Std error	p-value
intercept	-2.22	0.05	$<10^{-3}$
Pop	1.14	0.09	$<10^{-3}$
Density	-0.27	0.08	1.5×10^{-3}
Precip	-0.25	0.05	$<10^{-3}$
Rev/firm	0.13	0.06	0.0497
ISMT	0.35	0.07	$<10^{-3}$

because population is a driving force in the calculation of this use. Conversely, population density yielded a low coefficient ($\beta=0.09 \pm 0.01$).

Conversely, the industrial water use model showed low goodness-of-fit ($R_{adj}^2=0.12$

) and heteroscedasticity (controlled via robust standard errors). Despite the low variance explained, the results offer preliminary insights: (1) Negative associations with density ($\beta = -0.10 \pm 0.02$) and precipitation ($\beta = -0.08 \pm 0.01$) corroborate that industries favor low-density locations and utilize alternative water sources in arid zones; (2) While population and Rev/firm showed positive coefficients, the latter lost significance in the final model ($p=0.077$). The low fit suggests that industrial water variability is dominated by unobserved factors not captured by available commune-level socioeconomic datasets.

4.4 Urban Scaling

Figure 3 illustrates the logarithmic scaling relationships between commune size and water consumption sub-sectors DW, Industrial, and the sum of both (NLWU). The superlinear scaling of DW (Fig. 3A) indicates an increasing return of water consumption as the commu-

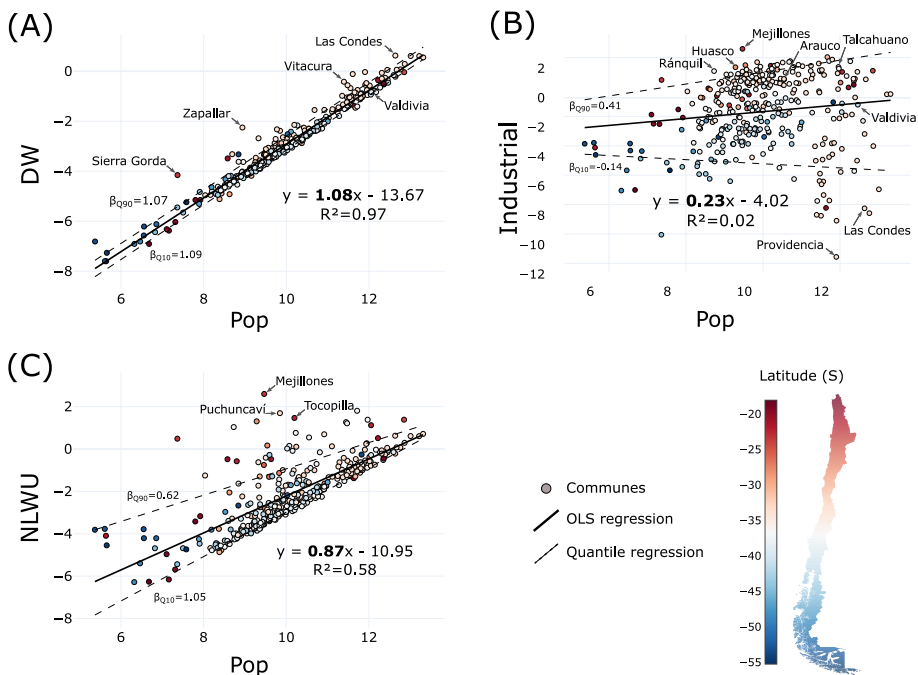


Fig. 3 Log—log urban scaling relationships between population and consumptive water use for Chilean communes in 2017. Each data point represents a commune ($n = 343$) colored by its latitude, and selected communes are labelled for illustrative purposes. Water use models are presented as (A) DW, (B) Industrial, and (C) both DW and Industrial aggregated. Each model is fitted using OLS regression (solid black line). The dashed lines in all panels indicate the trend fitted using Quantile Regression (β_{Q10} and β_{Q90}). Note: Communes with zero industrial water use ($n = 13$) were excluded from panel (B) for visualization purposes

nal population increases. Industrial water consumption shows a non-significant scaling relationship (Fig. 3B). Figure 3C shows the sublinear scaling with population size of NLWU.

While this cross-sectional analysis indicates that the sum of DW and Industrial water use scales sublinearly with size ($\beta=0.87 \pm 0.04$), quantile regression reveals heterogeneous regimes: communes with low water use relative to their size (Q10) scale near-proportionally ($\beta=1.08 \pm 0.011$), whereas high-intensity users (Q90) exhibit strong economies of scale ($\beta_{Q90}=0.62 \pm 0.13$) (Fig. 3C).

Disaggregating the sectors reveals divergent behaviors. DW exhibits a consistent super-linear trend ($\beta=1.08 \pm 0.01$), while industrial water use shows no significant scaling relationship ($R^2=0.02$) (Fig. 3A and B). We formally tested whether the scaling exponent differed significantly from linearity ($H_0: \beta=1$) using Wald tests with heteroskedasticity-robust standard errors (HC3). The estimated exponent resulted significantly greater than 1 in the DW model ($p(H_0: \beta=1) < 0.001$), confirming a superlinear scaling regime. The difference between NLWU and DW population elasticities is statistically significant ($t = -4.89$, $p < 10^{-4}$), confirming that while residential demand drives disproportionate growth in larger cities, industrial use attenuates this relationship. Consequently, subsequent scaling analyses focus exclusively on DW, where the population dependency is theoretically robust.

To identify structural outliers, we measured the Euclidean distance from the observed values to the scaling trendline. The communes exhibiting the largest deviations are predominantly located in the country's arid northern macrozone (e.g., Mejillones, Tocopilla). These outliers share a distinct functional profile: they are low-density territories hosting water-intensive strategic industries—specifically mining and energy generation—rather than large residential populations.

4.5 Scaling Exponents by Population Size and Urbanization Type

DW scales superlinearly with population size in the cross sectional model for all communes with a 95% CI lower bound of 1.05 (Fig. 4A). Stratification by population size reveals distinct trajectories between size classes. Larger communes ($> 20,000$ inhabitants) exhibited consistently higher exponents than smaller communes. Smaller communes ($\leq 20,000$ inhabitants) showed a downward drift in point estimates, with the confidence intervals frequently encompassing $\beta < 1$ — indicating that, for this size class, the scaling behavior is not statistically distinguishable from linearity or sublinearity. This pattern is consistent with greater heterogeneity of consumption in low-population areas, where local variation in density or land use may obscure any size-dependent scaling signal. When narrowing the analysis to strictly urban populations (Fig. 4B), the pattern of higher exponents in more populous communes persists.

5 Discussion

Our analysis of water use across Chilean communes reveals distinct spatial and socioeconomic patterns, offering critical insights into how metabolic performance varies in rapidly urbanizing environments, in terms of resource efficiency. By distinguishing between residential and industrial demands, we identify scaling patterns (e.g., $\text{Pop} \rightarrow \text{DW}$, $\beta > 1$) that challenge traditional urban sustainability frameworks.

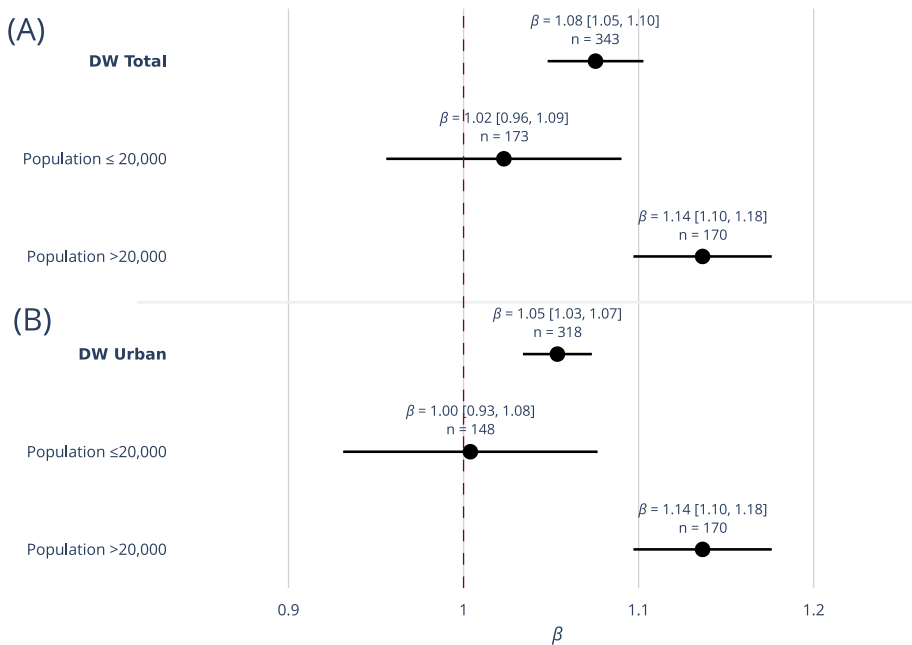


Fig. 4 Cross-sectional scaling exponents (β) for the relationship between drinking water (DW) and population, estimated by OLS for the year 2017. Point estimates and 95% confidence intervals are shown for three nested partitions of communes. **(A)** Total DW against total communal population: all 343 continental communes ($n=343$), communes $\leq 20,000$ inhabitants ($n=173$), and communes $> 20,000$ inhabitants ($n=170$). **(B)** Urban DW against urban communal population (excluding communes with zero urban DW or population): all 318 urban communes, urban communes $\leq 20,000$ inhabitants ($n=148$), and urban communes $> 20,000$ inhabitants ($n=170$). The vertical dashed line marks $\beta = 1$ (isometric scaling). Wald tests of $H_0 : \beta = 1$ for all six estimates are reported in Supplementary Table S2

5.1 The Superlinear Trap: Urban Scaling and Resource Inefficiency

The relationship between urban population and domestic water use exhibits superlinear scaling ($\beta > 1$) in the 2017 cross-section, with the pattern strongest among populous communes (Fig. 4). This finding challenges the theoretical expectation from UST, which proposes that basic infrastructural services should scale linearly ($\beta \approx 1$) or sublinearly with city size (Betencourt et al. 2007; Ribeiro and Rybski 2023). The persistence of this superlinear pattern—particularly among the most populous communes—is consistent with diseconomies of scale in the water metabolism of larger Chilean cities, where per capita water use rises with city size rather than contracting as classical infrastructure scaling would predict (Lu et al. 2024).

It is important to note that because DW estimates are partially derived from population data, where population figures are used as forcing data (see section), these models should not be interpreted as predictive tools, but rather as analytical instruments for assessing the scaling coefficient itself. The significant variation in β based on the definition of urban population reiterates the definitional paradox identified by (Cottineau et al. 2016) and (Arcaute et al. 2015), emphasizing the urgent need to transition from administrative boundaries to functional definitions of urban systems (Dong et al. 2024; Sotomayor-Gómez and Samaniego 2020). The greater variability observed in the coefficients for smaller urban

communes is consistent with a dominant role of local, non-scaling factors at low population sizes (Fig. 4B).

While sublinear scaling reflects internal metabolic efficiencies associated with urban infrastructure or dynamics, these efficiencies do not necessarily translate into sustainability gains once absolute consumption, externalization of impacts, and social inequities are taken into account.

5.2 The Industrial Decoupling and the Headquarters Effect

Two distinct β values for DW appear in our analysis and refer to different quantities. The bivariate scaling exponent ($\beta=1.08 \pm 0.01$; Fig. 3A) measures the gross elasticity of DW with respect to population in a simple log–log regression, the conventional estimand in the urban scaling literature. The multivariate coefficient ($\beta=1.29 \pm 0.02$; Table 1) is the partial elasticity of population in the Cobb–Douglas specification that also conditions on density, precipitation, Rev/firm, and ISMT. Because density is negatively associated with DW ($\beta=-0.27$) and strongly correlated with population ($\rho=0.71$), conditioning on density absorbs the density-mediated efficiency, leaving the partial population effect higher than the gross one. Both values support the same qualitative conclusion of superlinear scaling ($\beta>1$) and lie within the range reported elsewhere for socioeconomic urban indicators (e.g., Bettencourt et al. (2007)). While population size reliably predicts domestic demand, our analysis confirms that it cannot fully explain total water consumption. For Industrial use, while the fit improves in the multivariate model ($R^2_{adj}=0.12$; Table 1) compared to the simple linear regression ($R^2=0.02$; Fig. 3B), the explanatory power remains low. This suggests that other, more representative variables are needed to adequately capture the underlying drivers. This shift highlights that more representative variables are required to adequately capture the underlying dynamics of productive development (sensu Keuschnigg, Mutgan, and Hedström 2019).

Crucially, we find that industrial water use is largely decoupled from city size ($\beta=0.13$) and communal wealth (non-significant variables). The stark contrast between similarly sized yet functionally distinct communes (e.g., Talcahuano vs. Providencia in Fig. 3C) highlights that industrial location is driven by resource proximity rather than urban agglomeration.

However, our attempt to model this using official socioeconomic data revealed a critical “fiscal-functional mismatch” Because economic data are based on corporate tax records, they reflect the location of company headquarters rather than the site of resource extraction. This creates a spatial disconnection, where communes with high recorded productivity (e.g., Las Condes) show low industrial water use, while major industrial centers (e.g., Arauco) record lower economic productivity despite high water consumption. This mismatch precludes the effective use of administrative tax data for spatially accurate environmental modeling and highlights a blind spot in current national statistics.

5.3 Spatial Inequity: Sacrifice Zones and the Geography of Wealth

The decoupling of industrial water use reveals a spatial pattern that can be related to environmental injustice. Larger industrial water consumption is concentrated in lower-density territories, often coinciding with areas socially designated as “sacrifice zones” (e.g., Mejillones, Tocopilla in the North). This clustering of water-intensive activity—with its associ-

ated environmental externalities—points to an inequitable spatial distribution of the costs of national production (Anbleyth-Evans et al. 2022).

The analysis of wealth (proxied by ISMT) reveals a divergent spatial dynamic. While positive β coefficients in the NLWU and DW models indicate that aggregate water use scales with socioeconomic status, the non-significant relationship in the Industrial model suggests a structural decoupling (Table 1). The rise in aggregate water use is driven primarily by elevated domestic consumption in affluent residential zones. Conversely, major industrial consumers are predominantly located in communes with moderate-to-low ISMT scores (e.g., Coronel, Huasco, Ránquil). This reinforces a spatial separation of burden and benefit, wherein wealthier communes (e.g., Vitacura, Providencia) function as residential or service hubs, effectively ‘offshoring’ the physical demands of industrial metabolism to lower-density, less affluent territories.

Furthermore, industrial sourcing appears decoupled from local hydrology. Major extractive activities, particularly copper mining, are concentrated in the arid north, while low population density limits industrial growth in the water-rich south. Consequently, industrial water sourcing is driven by geologic availability rather than climatic suitability, necessitating complex technological interventions (e.g., desalination) that further complicate water management across Chile’s latitudinal gradient.

5.4 Implications for Planning in Complex Systems

The complexity of this urban-productive system forces us to adopt a perspective rooted in Complex Adaptive Systems and Planetary Boundaries. The relationship between urbanization and sustainability is non-linear; as Wei et al. (2025) suggests, while initial urbanization improves infrastructure, later stages often exacerbate environmental conflicts. Adopting a global boundaries approach is essential for delineating the constraints governing sustainable water use. However, this perspective could underestimate the complexity of hydrogeological processes and local tipping points (e.g., water availability in a basin).

A critical manifestation of this complexity is the “paradox of density” identified in our results: population density is positively associated with domestic water use ($\beta=0.09$) but negatively associated with total water use ($\beta=-0.27$). This implies that low-density industrial communes dedicate proportionally larger territories to water-intensive activities. For territorial planning, this offers a vital insight: strategies aimed at densification must be balanced against the risks of industrial land allocation in low-density zones, which drives total resource consumption.

This spatial disconnection poses a challenge for governance frameworks. While concepts such as carrying capacity and ecological flows address local limits, they often fail to link water exceedance to its underlying drivers (Bunsen et al. 2021). Our work quantifies this metabolic rift, showing that the communes with the highest absolute consumption (e.g., Mejillones, Tocopilla) are industrial centers rather than the most populous urban areas (Fig. 3). Furthermore, attributing spatially explicit water use entails uncertainty, as administrative boundaries rarely align with hydrological units. Some statistical procedures may also introduce bias. Stepwise regression, for instance, tends to overfit noise in the data, a risk that is amplified in cross-sectional studies owing to the absence of temporal replication here.

These combined mismatches—scale (administrative vs. basin), functional (extraction vs. headquarters), and spatial—reveal that relying on aggregate per capita indicators can

perpetuate structural injustices (Keuschnigg, Mutgan, and Hedström 2019). Simple metrics like SDG targets often obscure the true environmental impact of production in specific zones (Berger et al. 2023).

Sustainable water governance requires a multi-scale approach that traces these flows. A densely populated city with low local agricultural water use raises the fundamental question: Where does its metabolic support come from? Establishing sustainable limits requires addressing these cross-scale allocations—from local basins to global trade. By understanding why some communes achieve proportionally lower water use, we can generate actionable knowledge to navigate the challenges of water scarcity and move toward operating within the Earth System’s regenerative capacity.

5.5 Study Limitations

Several methodological limitations should be acknowledged. First, drinking water estimates from the CR2WU database are constructed using population as one of several forcing variables (see §2.1.1 and Supplementary §1). Although the sector-specific consumption rates K_s vary substantially across communes (Supplementary Fig. S1), introducing genuine empirical variation that the scaling regression captures, the partial reliance on population data implies that the reported β values reflect both real urban scaling and the structure of the data reconstruction. They should therefore be interpreted as descriptive scaling coefficients rather than as causal estimates, and as conservative upper bounds on the strength of the population-DW relationship. Second, communal-level socioeconomic indicators are census-derived and represent a single cross-Sect. (2017); we cannot capture inter-annual dynamics in productive structure or wealth that may modulate water demand. Third, stepwise selection based on AIC is sensitive to noise in cross-sectional data and may favor predictors that happen to correlate with the response in this particular sample; we mitigate this by anchoring variable inclusion in theoretical considerations (UST, Cobb–Douglas) and by reporting VIF iterations (Supplementary §3.3). Fourth, administrative communal boundaries rarely coincide with hydrological units, so attributions of water use to specific basins should be interpreted with caution. Finally, the CR2WU database holds sector-specific consumption rates K_s constant across the time series for each commune. This precludes a meaningful longitudinal analysis of β from these data; we have therefore restricted the scaling analysis to a single cross-Section (2017) where all inputs are independently sampled. These limitations do not undermine the main findings — superlinear DW scaling, industrial decoupling, and the spatial inequity of metabolic burden — but they define the inferential scope of this analysis. We offer it as an empirical synthesis to inform future research, rather than as a basis for direct prediction or policy targeting.

6 Conclusion

This research examined water use in Chile by jointly analyzing multiple types of water consumption alongside a broad set of demographic, environmental, and socioeconomic variables at the communal level. Disaggregating total consumptive water use revealed marked differences in how distinct forms of water consumption relate to urban size, location, and socio-economic organization.

Water use associated with land use (LWU) exhibited no consistent relationship with the examined communal variables, highlighting the difficulty of capturing the drivers of this type of consumption at the communal scale. In contrast, drinking water and industrial sectors aggregated (NLWU) showed a clearer and more structured relationship with urban characteristics. A parsimonious subset of five variables explained a substantial share of its variation within a linearized Cobb–Douglas framework, yielding a robust model fit ($R^2=0.672$). The industrial sector, analyzed in isolation, while statistically significant, displayed limited explanatory power ($R^2=0.12$, $F=12.68$, $p<10^{-3}$), underscoring the persistent challenge of identifying key drivers of productive water use across Chilean communes.

Drinking water use exhibited superlinear scaling with population size, indicating that larger communes intensify per capita domestic water demand. This scaling relationship was not homogeneous across the urban system: exponents varied with population size class and between total and strictly urban definitions of population, revealing a superlinear pattern that is stronger in larger and more urbanized contexts.

Taken together, these findings indicate that scaling exponents provide insight into the internal organization of urban water metabolism, understood here as the spatial distribution of population, the configuration of water infrastructure, and the concentration of socio-economic activities that shape water flows across communes. At the same time, they highlight the limits of efficiency-based interpretations when used in isolation to assess sustainability outcomes. The observed scaling patterns and spatial inequalities suggest that contemporary urban water dynamics are not solely driven by population size or efficiency gains, but are shaped by sectoral decoupling — particularly between industrial and residential demand — that redistributes environmental burdens beyond city boundaries. From a planning perspective, addressing urban water sustainability therefore requires moving beyond sector-specific or purely technological solutions toward approaches that explicitly conceptualize cities as socio-ecological systems embedded within biophysical constraints and characterized by uneven socio-environmental outcomes.

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Data Availability https://github.com/horaciosamaniego/WaterMetabolism_Chile_2026/

Declarations

Competing Interests The authors declare no competing interests.

References

- Alberti M (2004) Ecological resilience in urban ecosystems: linking urban patterns to human and ecological functions. *Urban Ecosyst* 7(3):241–265. <https://doi.org/10.1023/B:UECO.0000044038.90173.c6>
- Alvarez-Garretón C, Lefort I, Marinao R, Barria P (2023) Protecting environmental flows to achieve long-term water security. *J Environ Manage* 328:116914. <https://doi.org/10.1016/j.jenvman.2022.116914>
- Alves L, Ribeiro EK, Lenzi M (2013) Distance to the scaling law: a useful approach for unveiling relationships between crime and urban metrics. *PLoS One*. <https://doi.org/10.1371/journal.pone.0069580>
- Anbley-Evans J, García Cegarra A (2022) Toxic violence in marine sacrificial zones: developing blue justice through marine democracy in Chile. *Environ Plan C Polit Space* 40(7):1492–1514
- Arcaute E, Hatna E (2015) Constructing cities, deconstructing scaling laws. *J R Soc Interface* 12(102):20140745. <https://doi.org/10.1098/rsif.2014.0745>
- Babel MS, Das Gupta A, Pradhan P (2007) A multivariate econometric approach for domestic water demand modeling: an application to Kathmandu, Nepal. *Water Resour Manage* 21(3):573–589. <https://doi.org/10.1007/s11269-006-9030-6>
- Central Bank of Chile (2024) Cuentas Nacionales de Chile 2018–2024
- Batty M (2013) A theory of city size. *Science* 340(6139):1418–1419. <https://doi.org/10.1126/science.1239870>
- Berger L, Adam DH, Pivo G (2023) Orienteering the Landscape of Urban Water Sustainability Indicators. *Environ Sustain Indic* 17:100207. <https://doi.org/10.1016/j.indic.2022.100207>
- Bettencourt LMA (2013) The origins of scaling in cities. *Science* 340(6139):1438–41. <https://doi.org/10.1126/science.1235823>
- Bettencourt LMA, Helbing D (2007) Growth, innovation, scaling, and the pace of life in cities. *Proc Natl Acad Sci U S A* 104:7301–7306. <https://doi.org/10.1073/pnas.0610172104>
- Boisier JP (2023) CR2MET: a high-resolution precipitation and temperature dataset for the period 1960–2021 in Continental Chile. Dataset, Version 2:5. <https://doi.org/10.5281/zenodo.7529682>
- Boisier JP, Marinao R, Álvarez-Garretón C (2024) CR2WU: water use dataset for the period 1950–2020 in Continental Chile. Zenodo. <https://doi.org/10.5281/zenodo.13324235>
- Boisier JP, Álvarez-Garretón C, Marinao R, Galleguillos M (2025) Increasing water stress in Chile revealed by novel datasets of water availability, land use and water use. *Hydrol Earth Syst Sci* 29(20):5185–5212. <https://doi.org/10.5194/hess-29-5185-2025>
- Brondízio ES, Settele J, Díaz S, Ngo HT (2019) Global assessment report of the Intergovernmental Science-Policy Platform on Biodiversity. IPBES secretariat
- Bunsen J, Berger M, Finkbeiner M (2021) Planetary boundaries for water – a review. *Ecol Indic* 121:107022. <https://doi.org/10.1016/j.ecolind.2020.107022>
- Burger JR, Allen JH, Brown WR, Burnside AD, Davidson TS, Fristoe MJ, Hamilton, Allen CD, Brown JH, Burnside WR, Davidson AD, Fristoe TS et al (2012) The macroecology of sustainability. *PLoS Biol*. <https://doi.org/10.1371/journal.pbio.1001345>
- Burger JR, Brown JW, Day TP, Flanagan, Roy (2019) The central role of energy in the urban transition: global challenges for sustainability. *BioPhys Econ Res Qual* 4(1):5. <https://doi.org/10.1007/s41247-019-0053-z>
- Burger JR, Okie IA, Hatton VP, Weinberger MS, Shrestha M, Liedtke TB, Be Tea, Okie JG, Hatton IA, Weinberger VP, Liedtke KJ et al (2022) Global city densities: re-examining urban scaling theory. *Front Conserv Sci* 3:1–10. <https://doi.org/10.3389/fcsc.2022.879934>
- CASEN (2017) Base de Datos Casen 2017 SPSS: Encuesta de Caracterización Socioeconómica Nacional.” Ministerio de Desarrollo Social y Familia; Dataset. <https://observatorio.ministeriodesarrollosocial.gob.cl/encuesta-casen-2017>
- Congreso Nacional de Chile (1981) Decreto con Fuerza de Ley N.º 1122: Fija texto del Código de Aguas. Ministerio de Justicia
- Cottineau C, Hatna E, Arcaute E (2016) Diverse cities or the systematic paradox of urban scaling laws. *Comput Environ Urban Syst*. <https://doi.org/10.1016/j.compenvurbysys.2016.04.006>
- D’Acci LS (ed) (2024) Urban scaling: allometry in urban studies and spatial science. Routledge, London. <https://doi.org/10.4324/9781003288312>
- Delgado-Ramos G (2015) Water and the political ecology of urban metabolism: The case of Mexico City. *J Pol Ecol* 22(1). <https://doi.org/10.2458/v22i1.21080>
- Dong L, Duarte F, Duranton G, Santi P, Barthelemy M, Batty M, Bettencourt Lea et al (2024) Defining a city — delineating urban areas using cell-phone data. *Nat Cities* 1(2):117–125. <https://doi.org/10.1038/s44284-023-00019-z>
- Elmqvist T, Fragkias M, Goodness J, Güneralp B, Marcotullio PJ, McDonald RI, Parell S (eds) (2013) Urbanization, biodiversity and ecosystem services: challenges and opportunities. Springer. <https://doi.org/10.1007/978-94-007-7088-1>

- Guetté A, Gaüzère P, Devictor V, Jiguet F, Godet L (2017) Measuring the synanthropy of species and communities to monitor the effects of urbanization on biodiversity. *Ecol Indic* 79:139–54. <https://doi.org/10.1016/j.ecolind.2017.04.018>
- Güneralp B, McDonald MF, Marcotullio, Seto (2013) Chapter 22 urbanization forecasts, effects on land use, biodiversity, and ecosystem services. In: Elmqvist T, Fragkias M, Goodness J, Güneralp B, Marcotullio PJ, McDonald RI, Parnell S (eds) *Urbanization, biodiversity and ecosystem services: challenges and opportunities*. Springer, pp 437–452. https://doi.org/10.1007/978-94-007-7088-1_22
- Hijmans J (2025) Terra: spatial data analysis. <https://github.com/rspatial/terra>
- INE (2018) Síntesis de Resultados Censo 2017. Instituto Nacional de Estadísticas. <http://resultados.censo2017.cl/>
- INE (2021) Proyecciones de Población. Dataset; Instituto Nacional de Estadísticas. <https://www.ine.gob.cl/estadisticas/sociales/demografia-y-vitales/proyecciones-de-poblacion>
- INE (2025) Censo 2024 Resultados Viviendas y Servicios Básicos. Instituto Nacional de Estadísticas. <https://censo2024.ine.gob.cl/documentacion/>
- Keuschnigg M (2019a) Scaling trajectories of cities. *Proc Natl Acad Sci U S A* 116(28):13759–13761. <https://doi.org/10.1073/pnas.1906258116>
- Keuschnigg M (2019b) Urban scaling and the regional divide. *Sci Adv* 5(1):eaav0042. <https://doi.org/10.1126/sciadv.aav0042>
- Koenker R, Bassett G (1978) Regression quantiles. *Econometrica* 46(1):33–50. <https://doi.org/10.2307/1913643>
- Liu Z, Lang W, Liu Y (2025) Unveiling local dynamics of Chinese cities: a multidimensional scale-adjusted analysis. *Habitat Int* 166:103596. <https://doi.org/10.1016/j.habitatint.2025.103596>
- Lobo J, Alberti M, Allen-Dumas M, Arcaute E, Barthelemy M, Bojorquez LA, Tapia SB et al (2020) Urban science: integrated theory from the first cities to sustainable metropolises. *SSRN Electron J*. <https://doi.org/10.2139/ssrn.3526940>
- Loreau M (2022) The ecological and societal consequences of Biodiversity Loss. In: Loreau M, Hector A, Isbell F (eds) 1st ed. Wiley. <https://doi.org/10.1002/9781119902911>
- Lu M, Jackson K (2024) Worldwide scaling of waste generation in urban systems. *Nat Cities* 1(2):126–135. <https://doi.org/10.1038/s44284-023-00021-5>
- Marquet PA, Gaxiola A, Alaniz A, Etcheberry G, Jara V, Menares L (2022) Las Tres Brechas Del Desarrollo Sostenible y El Cierre de La Brecha Ambiental En Chile: Oportunidades Para Una Recuperación Pospandemia Más Sostenible y Con Bajas Emisiones de Carbono En América Latina y El Caribe. *LC/TS.2022/35*. Santiago: Comisión Económica para América Latina y el Caribe (CEPAL). <https://www.cepal.org/es/publicaciones/47883-tres-brechas-desarrollo-sostenible-cierre-la-brecha-ambiental-chile>
- MINDESO (1992) Cuadros Estadísticos Demografía Casen. Dataset; Ministerio de Desarrollo Social y Familia. <https://observatorio.ministeriodesarrollosocial.gob.cl/encuesta-casen-1992>
- MINVU (2009) Circular DDU-227 Planificación Urbana, Formulación y Contenidos Plan Regulador Comunal.” Chile: Ministerio de la Vivienda y Urbanismo. <https://www.minvu.gob.cl/wp-content/uploads/2019/06/DDU-227.pdf>
- Morrone JJ (2015) Biogeographical regionalisation of the Andean region. *Zootaxa*. <https://doi.org/10.11646/zootaxa.3936.2.3>
- Nations, United (2021) The United Nations world water development report: Valuing water. Edited by UN Water. Paris: UNESCO
- Newbold T, Hudson LN, Hill SLL, Contu S, Lysenko I, Senior RA, Börger Lea et al (2015) Global effects of land use on local terrestrial biodiversity. *Nature* 520(7545):45–50. <https://doi.org/10.1038/nature14324>
- OCUC (2019) Índice Socio Material Territorial ISMT.” Observatorio de Ciudades UC. <http://observatoriodeciudades.com/portfolio/ismt/>
- Peng J, Liu Q, Liu Y (2020) Quantifying spatial morphology and connectivity of urban heat islands in a megacity: a radius approach. *Sci Total Environ* 714:136792, 136792. <https://doi.org/10.1016/j.scitotenv.2020.136792>
- Quinn G (2002) *Experimental design and data analysis for biologists*. Cambridge University Press
- Ravinandrasana VP (2025) The first emergence of unprecedented global water scarcity in the Anthropocene. *Nat Commun* 16(1):8281, 8281. <https://doi.org/10.1038/s41467-025-63784-6>
- Ribeiro FL, Rybski D (2023) Mathematical models to explain the origin of urban scaling laws. *Phys Rep* 1012:1–39. <https://doi.org/10.1016/j.physrep.2023.02.002>
- Ribeiro FL, Meirelles J, Netto CRN (2020) On the relation between transversal and longitudinal scaling in cities. *PLoS ONE*. <https://doi.org/10.1371/journal.pone.0233003>
- Rockström J, Steffen W, Noone K, Persson FS, Chapin, Lambin E, Lenton TM, Scheffer M et al. (2009) Planetary boundaries: exploring the safe operating space for humanity. *Ecol Soc* 14(2). <https://doi.org/10.5751/ES-03180-140232>

- Sarkar S (2019) Urban scaling and the geographic concentration of inequalities by city size. *Environ Plan B Urban Anal City Sci* 46(9):1627–1644. <https://doi.org/10.1177/2399808318766070>
- SII (2024) Estadística de Empresas Por Comuna. Servicio de Impuestos Internos; Dataset.
- Sotomayor-Gómez B, Samaniego H (2020) City limits in the age of smartphones and urban scaling. *Comput Environ Urban Syst* 79:101423, 101423. <https://doi.org/10.1016/j.compenvurbsys.2019.101423>
- UNGA, United Nations General Assembly (2010) The Human Right to Water and Sanitation.” United Nations {Resolution} A/RES/64/292. UN. <https://digitallibrary.un.org/record/687002>
- UN-Habitat, United Nations Human Settlements Programme (2022) World cities report 2022: envisaging the future of cities. United Nations Human Settlements Programme, Nairobi, Kenya
- UN-Water (2013) Water security and the global water agenda: a UN-Water Analytical Brief. Hamilton, Ont.: United Nations University - Institute for Water, Environment; Health
- Valdés-Pineda R, Pizarro R, García-Chevesich P, Valdés CO, Olivares C, Vera M, Balocchi Fea, Valdés JB et al (2014) Water governance in Chile: availability, management and climate change. *J Hydrol* 519:2538–2567. <https://doi.org/10.1016/j.jhydrol.2014.04.016>
- Van Vuuren DP, Doelman JC, Tagomori IS, Beusen AHW, Cornell SE, Röckstrom J, Schipper Aea et al (2025) Exploring pathways for world development within planetary boundaries. *Nature* 641(8064):910–16. <https://doi.org/10.1038/s41586-025-08928-w>
- Wei X, Zhao X, Wang P, Gao P, Zhong F (2025) The impact of urbanization on the interactions among the sustainable development goals: dynamic evolution and nonlinear changes in underdeveloped mountainous areas. *Ecol Indic* 179:114228, 114228. <https://doi.org/10.1016/j.ecolind.2025.114228>

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